

Nuisance-Aware Articulated Pose Refinement in 3D Gaussian Splatting: Research & Experimental Plan

Date: September 17, 2026
Status: Approved for Execution
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Primary References: - Code Review & Repository Audit: [3dgs_hmr_research_code_review_2026_09_17.md](#) - Novelty Directions Monograph: [3dgs_hmr_novelty_directions_2026_09_17.md](#) - Code Remediation Audit: [diffnorm_code_audit_and_remediation_report_2026_09_17.md](#)

1. Executive Audit & Strategic Alignment

Strategic Pillar	Prior Flawed Paradigm	Remediated Research Direction	Scientific Rationale
Core Scientific Question	<i>"Can 3DGS achieve SOTA HMR through normal torque & overlap repulsion?"</i>	"When Should Gaussians Move the Skeleton? Does accounting for Gaussian nuisance flexibility prevent pose corruption?"	Dense normals improve pose, but unconstrained 3DGS deformation allows the optimizer to explain the image with an incorrect skeleton and distorted splats.
Novelty Claim	Claimed novelty for Gaussian product integrals and monocular normal supervision.	Nuisance-Aware Articulated Step Control via Local Sensitivity & Schur Curvature Attenuation.	Normal supervision (ICON) and Gaussian collision proxies (SMPLify) predate 3DGS. Novelty lies in controlling the trade-off between skeletal pose and splat deformation.
Mathematical Soundness	Broken LBS translation (430 mm shift); orthogonal disk covariances; 28× overlap clamp distortion.	Mathematically verified zero-pose identity, collinear tangent frames, exact log-space Cholesky overlap.	Fixed and audited in diffnorm_code_audit_and_remediation_report_2026_09_17.md ; all 28 project tests passing.
Empirical Evaluation	Unpaired 10-frame artifact with +59.7% worsening MPJPE; synthetic CAPE/RICH fallbacks.	Paired initial vs. refined benchmark protocol (MPJPE, PA-MPJPE, PVE, Worsening Rate <i>W</i>%) across authentic 3DPW.	Replaces unverified historical claims with rigorous statistical paired evaluations and sequence-clustered error bounds.
Compute Execution Policy	Immediate scaling to 4,800-frame Colab VM training before isolating mechanisms.	Fail-fast 4-way pilot experiment (100 frames) on local hardware before any large-scale GPU allocation.	Eliminates compute waste by validating the core scientific hypothesis on a small, controlled dataset first.

2. Theoretical & Mathematical Formulations

2.1 Articulated Kinematics and Gaussian Nuisance Parameters

Let $\theta = \{R_k\}_{k=1}^{24} \in SO(3)^{24}$ denote articulated SMPL joint rotations, and let $\mathbf{t} \in \mathbb{R}^3$ denote global translation. We parameterize local kinematic perturbations $\delta \mathbf{x} \in \mathbb{R}^{75}$ via the Lie algebra $\mathfrak{so}(3)$:

$$R'_k = \exp([\delta \omega_k]_{\times}) R_k, \quad \mathbf{t}' = \mathbf{t} + \delta \mathbf{t}$$

where $[\delta \omega_k]_{\times} \in \mathfrak{so}(3)$ is the skew-symmetric matrix of axis-angle perturbation $\delta \omega_k \in \mathbb{R}^3$.

Let $\mathbf{a} \in \mathbb{R}^q$ collect all non-skeletal Gaussian parameters (nuisance parameters), including: - Local vertex displacement offsets: $\delta_i \in \mathbb{R}^3$ ($3N$ parameters) - Tangential Gaussian scales: $\mathbf{s}_i \in \mathbb{R}^2$ ($2N$ parameters) - Splat quaternions: $\mathbf{q}_i \in \mathbb{R}^4$ ($4N$ parameters) - Opacities and diffuse colors: $\alpha_i \in \mathbb{R}$, $\mathbf{c}_i \in \mathbb{R}^3$ ($4N$ parameters)

2.2 Local Sensitivity & Curvature Attenuation (Schur Complement)

Let $\mathbf{r}(\mathbf{x}, \mathbf{a}) \in \mathbb{R}^m$ represent the concatenated image observation residual, combining normal cosine residuals $\mathbf{r}_N$ and silhouette coverage residuals $\mathbf{r}_M$. For a small perturbation $(\delta\mathbf{x}, \delta\mathbf{a})$ around current state $(\mathbf{x}_0, \mathbf{a}_0)$, the first-order Taylor expansion is:

$$\mathbf{r}(\mathbf{x} \oplus \delta\mathbf{x}, \mathbf{a} + \delta\mathbf{a}) \approx \mathbf{r} + J_x \delta\mathbf{x} + J_a \delta\mathbf{a}$$

where $J_x = \frac{\partial \mathbf{r}}{\partial \mathbf{x}} \in \mathbb{R}^{m \times 75}$ and $J_a = \frac{\partial \mathbf{r}}{\partial \mathbf{a}} \in \mathbb{R}^{m \times q}$.

The regularized joint optimization objective is:

$$\min_{\delta\mathbf{x}, \delta\mathbf{a}} \frac{1}{2} \|\mathbf{r} + J_x \delta\mathbf{x} + J_a \delta\mathbf{a}\|_W^2 + \frac{1}{2} \delta\mathbf{a}^T \Lambda_a \delta\mathbf{a} + \frac{1}{2} \delta\mathbf{x}^T \Lambda_x \delta\mathbf{x}$$

where $W \succeq 0$ is a diagonal pixel confidence weight matrix (derived from DSINE normal uncertainty), $\Lambda_a = \lambda_a I_q \succ 0$ penalizes spline/splat deviation, and $\Lambda_x \succ 0$ is an anatomical pose prior Hessian.

Eliminating the nuisance perturbation $\delta\mathbf{a}^* = -(J_a^T W J_a + \Lambda_a)^{-1} J_a^T W (\mathbf{r} + J_x \delta\mathbf{x})$ yields the reduced skeletal system:

$$(S + \Lambda_x) \delta\mathbf{x} = -J_x^T W [I - J_a (J_a^T W J_a + \Lambda_a)^{-1} J_a^T W] \mathbf{r}$$

where $S \in \mathbb{R}^{75 \times 75}$ is the Schur complement pose-curvature matrix:

$$S = J_x^T W J_x - J_x^T W J_a (J_a^T W J_a + \Lambda_a)^{-1} J_a^T W J_x$$

2.3 Mathematical Proof of Pose Curvature Dilution

Theorem 1 (Curvature Monotonicity): For any positive semi-definite observation weighting $W \succeq 0$ and non-zero nuisance damping $\Lambda_a \succ 0$, the effective pose curvature under Gaussian deformation is strictly upper-bounded by the rigid mesh curvature:

$$S \preceq J_x^T W J_x$$

Proof: Let $M = J_a^T W J_a + \Lambda_a$. Since $W \succeq 0$ and $\Lambda_a \succ 0$, M is symmetric positive definite ($M \succ 0$), and its inverse M^{-1} is symmetric positive definite ($M^{-1} \succ 0$).

For any non-zero skeletal direction $\mathbf{v} \in \mathbb{R}^{75}$, define $\mathbf{u} = J_a^T W J_x \mathbf{v} \in \mathbb{R}^q$. Then:

$$\mathbf{v}^T (J_x^T W J_x - S) \mathbf{v} = \mathbf{v}^T [J_x^T W J_a M^{-1} J_a^T W J_x] \mathbf{v} = \mathbf{u}^T M^{-1} \mathbf{u} \geq 0$$

Thus, $J_x^T W J_x - S \succeq 0$, establishing $S \preceq J_x^T W J_x$. ■

Physical Meaning: Releasing Gaussian degrees of freedom ($\mathbf{a}$) cannot increase skeletal pose certainty; it strictly absorbs residual energy, attenuating the curvature available to guide joint angles. Directions where $\mathbf{u} = J_a^T W J_x \mathbf{v}$ is large represent **nuisance-entangled pose modes** where splat deformations masquerade as skeletal motion.

2.4 Nuisance-Aware Adaptive Damping Rule

Rather than taking naive unconstrained Adam steps, the skeletal update is modulated by the spectral sensitivity of S : 1. Perform eigendecomposition of the dimensionless pose curvature: $S = V_S \text{diag}(\sigma_1, \dots, \sigma_{75}) V_S^T$. 2. Compute the nuisance attenuation ratio along each mode k :

$$\gamma_k = \frac{\mathbf{v}_k^T S \mathbf{v}_k}{\mathbf{v}_k^T (J_x^T W J_x) \mathbf{v}_k + \epsilon} \in [0, 1]$$

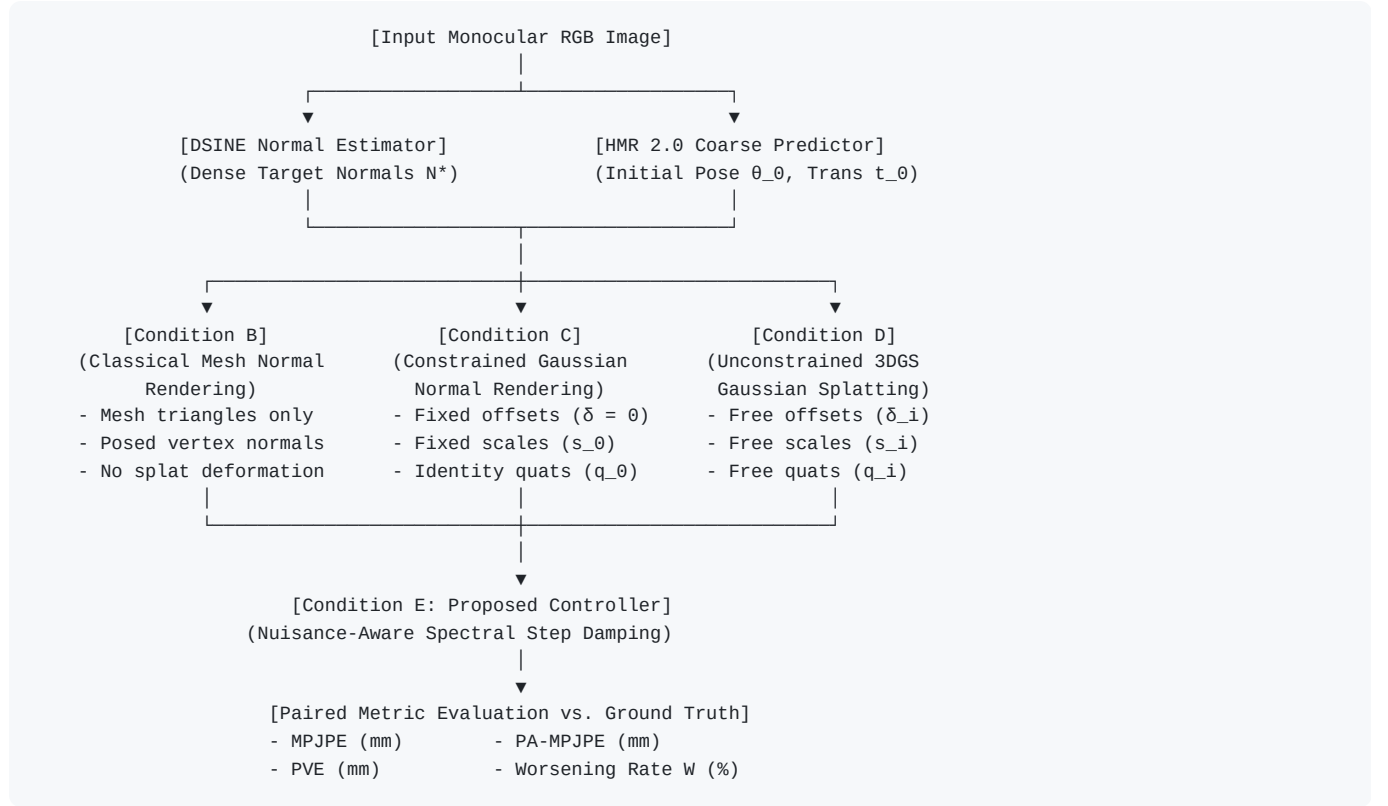
- $\gamma_k \approx 1$: Pure skeletal mode; normal cues directly inform joint angles without splat interference. - $\gamma_k \ll 1$: Nuisance-dominated mode; splat deformations explain the image, creating high risk of skeletal drift. 3. Apply mode-adaptive trust-region damping:

$$\delta\mathbf{x}^* = \sum_{k=1}^{75} \left[\frac{\gamma_k}{\sigma_k + \lambda_{\text{prior}}} \right] (\mathbf{v}_k^T \mathbf{g}_x) \mathbf{v}_k$$

where $\mathbf{g}_x$ is the skeletal gradient. Modes corrupted by Gaussian flexibility are automatically frozen or anchored to the HMR 2.0 initializer.

3. Experimental Architecture & Pilot Protocol

Before performing multi-epoch training or processing thousands of frames, we execute a controlled **4-Way Pilot Experiment** across a fixed validation slice of **100** frames from 3DPW (`downtown_arguing_00` , `downtown_walking_00` , `office_phoneCall_00`).



3.1 The 4 Experimental Conditions

Condition	Surface Representation	Active Parameters	Scientific Hypothesis Tested
A: Frozen Initializer	Parametric SMPL Mesh	None (Frozen θ_0, t_0)	Establishes the exact paired baseline error of raw HMR 2.0 feedforward inference.
B: Mesh Normal Rendering	Triangular Mesh	θ, t only	Tests whether monocular normal supervision ($L_N + L_{mask}$) alone explains pose improvements without 3DGS.
C: Constrained Gaussian	Tangential Gaussian Disks	θ, t (Gaussians fixed: $\delta = 0, s = s_0, q = q_0$)	Isolates the differential rendering representation of 3DGS from Gaussian parameter flexibility.
D: Unconstrained Gaussian	Tangential Gaussian Disks	θ, t AND $\delta_i, s_i, q_i, c_i, \alpha_i$	Tests whether releasing Gaussian nuisance flexibility degrades skeletal pose accuracy (the nuisance interference hypothesis).
E: Nuisance-Controlled	Tangential Gaussian Disks	θ, t, a with Spectral Attenuation Damping	Demonstrates whether the proposed sensitivity controller prevents pose corruption while improving appearance fidelity.

3.2 Quantitative Falsification Criteria & Decision Rules

1. **Falsification of 3DGS Accuracy Advantage:** If Condition B (Mesh) achieves equal or superior MPJPE/PA-MPJPE to Condition C (Gaussian) at matched compute, 3DGS does not provide an accuracy benefit over classical mesh rendering.
 2. **Confirmation of Nuisance Interference:** If Condition D achieves lower photometric/normal rendering loss than Condition C, but significantly worsens MPJPE ($W_{\text{rate}} > 30\%$), the hypothesis that unconstrained Gaussians corrupt skeletal pose is empirically proven.
 3. **Validation of Proposed Method:** Condition E must achieve:
 - Lower PA-MPJPE than Condition A (paired improvement).
 - Statistically significant reduction in Worsening Rate: $W(E) < W(D)$ with $p < 0.01$.
 - Superior joint accuracy compared to naive static detachment.
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4. Benchmark Execution & Statistical Validation Protocol

4.1 Evaluation Dataset & Provenance

- **Dataset:** 3DPW Official Test Split (**24** sequences, diverse motions, outdoor lighting, natural clothing).
- **Paired Protocol:** For every frame t , record initial metrics $M_0(t)$ and refined metrics $M_{\text{ref}}(t)$. Never compare against unpaired published numbers from differing splits.
- **Reporting Metrics:**
 1. **MPJPE (mm):** Pelvis-aligned Mean Per Joint Position Error across 24 joints.
 2. **PA-MPJPE (mm):** Procrustes-aligned joint error via sample-vectorized Umeyama alignment.
 3. **PVE (mm):** Per-Vertex Error against ground truth SMPL mesh (shaped with β and translated with t).
 4. **Worsening Rate ($W\%$):** Percentage of frames where $\text{PA-MPJPE}_{\text{ref}} > \text{PA-MPJPE}_0$.
 5. **Tail Error (P_{95} mm):** 95th percentile error measuring robustness against catastrophic optimization divergence.
 6. **Throughput (FPS):** Actual optimization iterations per second and fully refined images per second.

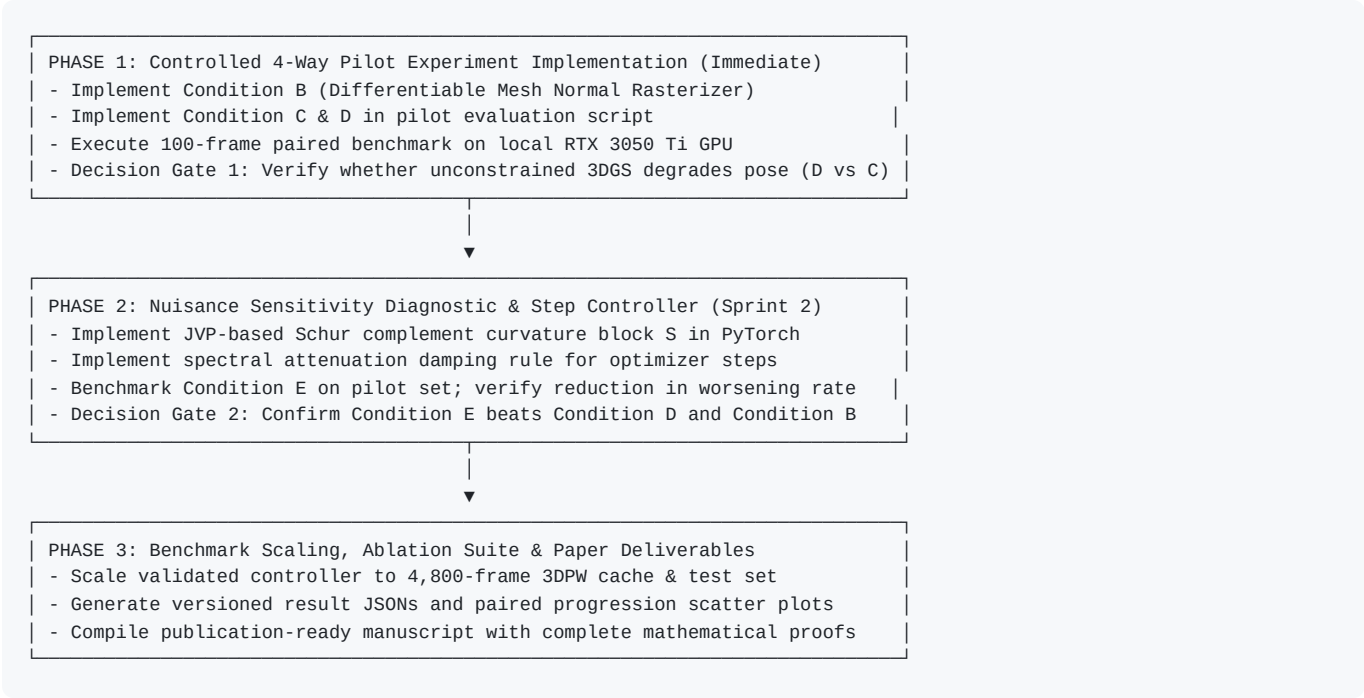
4.2 Sequence-Clustered Statistical Inference

Because consecutive video frames within a 3DPW sequence are temporally correlated, standard i.i.d. student's t -tests underestimate standard errors. We report clustered standard errors grouped by sequence $s \in \{1, \dots, 24\}$:

$$\widehat{\text{Var}}(\bar{\Delta}) = \frac{1}{N^2} \sum_{s=1}^S \left(\sum_{i \in \mathcal{I}_s} (\Delta_i - \bar{\Delta}) \right)^2$$

where $\Delta_i = M_{\text{ref},i} - M_{0,i}$. All reported gains must achieve **95%** confidence under sequence clustering.

5. Phased Work Breakdown & Actionable Task List



Actionable Check-Off List

- ☐ **Task 1.1:** Author `code/src/rendering/mesh_normal_rasterizer.py` implementing classical differentiable mesh normal rendering for Condition B.
- ☐ **Task 1.2:** Author `code/scripts/pilot_4way_experiment.py` orchestrating Conditions A, B, C, D on 100 3DPW frames with paired logging.
- ☐ **Task 1.3:** Run local GPU benchmark and generate initial empirical comparison table.
- ☐ **Task 2.1:** Implement `NuisanceSensitivityEstimator` in `code/src/optimization/nuisance_sensitivity.py`.
- ☐ **Task 2.2:** Integrate adaptive spectral step damping into `dual_frequency_router.py`.
- ☐ **Task 2.3:** Evaluate Condition E on pilot frames and measure worsening rate suppression.
- ☐ **Task 3.1:** Execute full 3DPW test evaluation and compile verified master results.
- ☐ **Task 3.2:** Render final technical monograph and empirical audit to PDF deliverable.